



Agentic SDLC Software Factory

Governed multi-agent software delivery from requirements to operations

[GitHub documentation](#)

[csdmichael/Foundry-Agentic-Workflow-SDLC-Docs](https://github.com/csdmichael/Foundry-Agentic-Workflow-SDLC-Docs)

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POC prepared collaboratively

The application in one view

A responsive Software Factory that combines specialized Foundry agents, explicit workflow control, and enterprise systems of record.

12 specialized agents

Requirements through Insights

3 execution policies

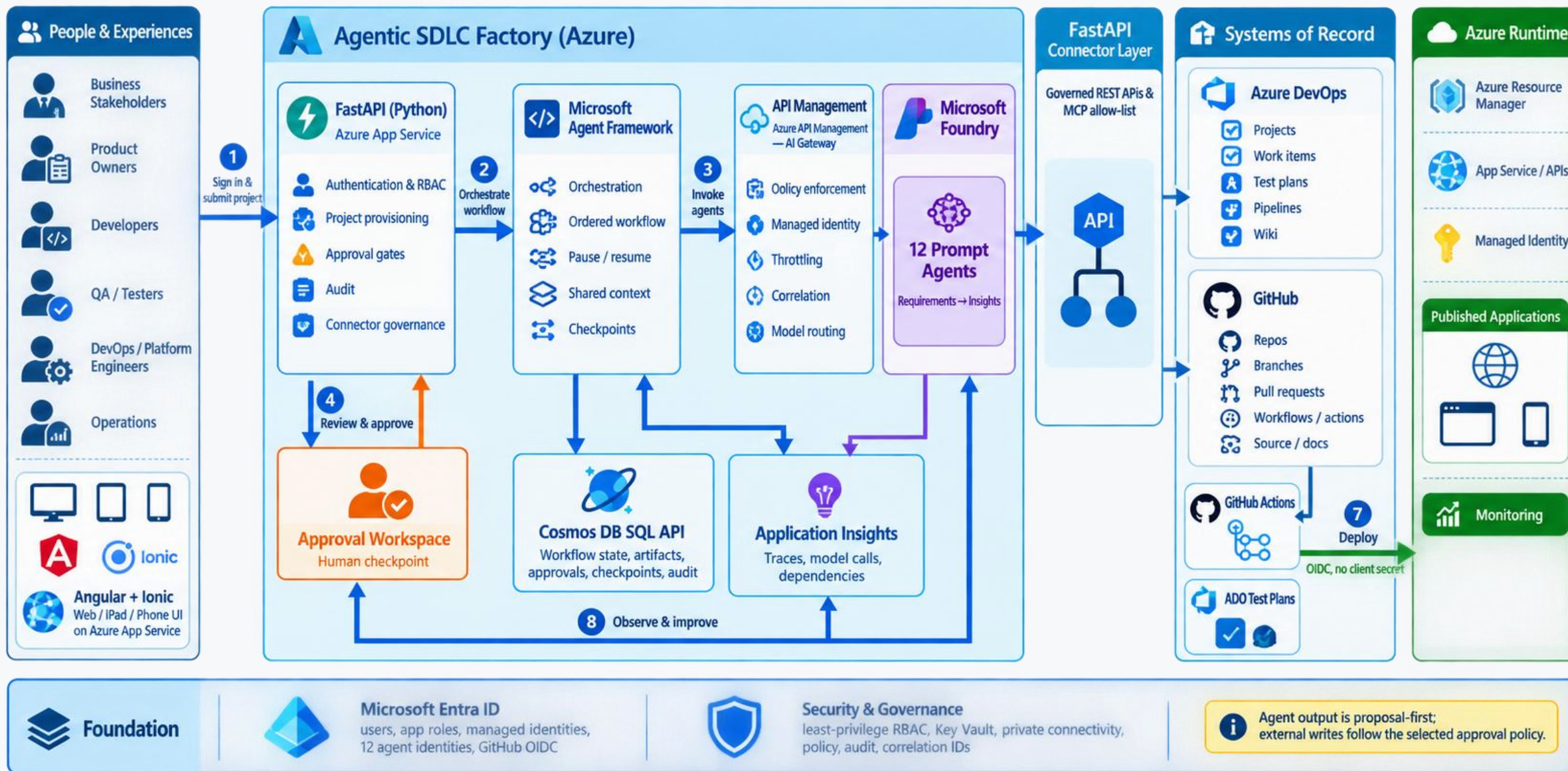
Autonomous, minimal review, human review

9 evidence-aware gates

Policy or named-human decisions

Durable execution

Checkpoint, resume, retry and audit



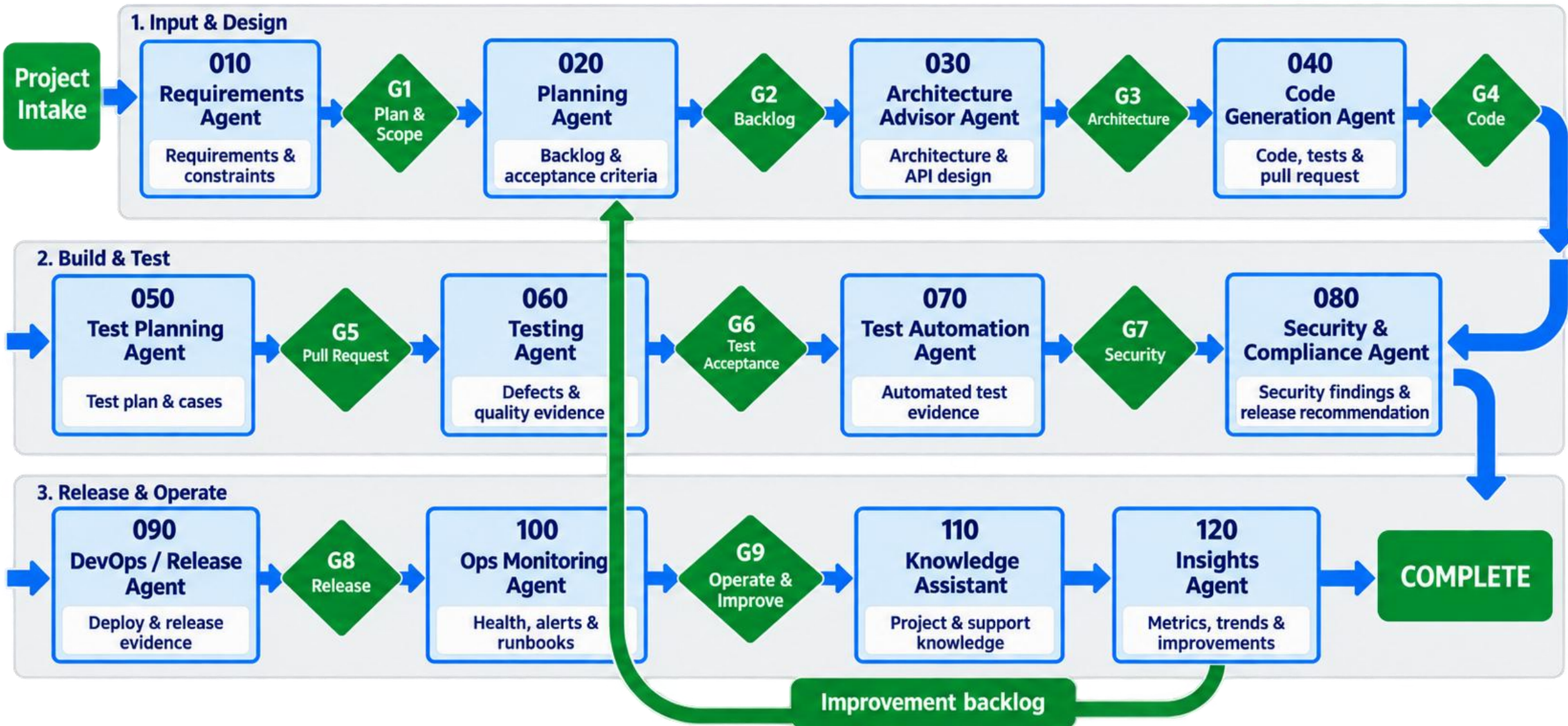
How the framework patterns map to this POC

The application primarily uses an explicit sequential graph, with policy branches, revision loops, and human checkpoints.

Sequential	Core SDLC progression	Guarantees stage order and evidence inheritance
Conditional	Approval policy and failure handling	Routes autonomous, minimal-review, human-review and error paths
Loop	Request changes and improvement	Returns reviewer feedback to the owning agent
Concurrent	Project sets and selected future optimizations	Runs independent projects or safe read-only analysis in parallel
Handoff / Group chat / Magentic	Optional advanced collaboration	Useful where dynamic delegation or debate adds measurable value

Agentic SDLC — Fully Autonomous Workflow

 **AUTOMATIC** —
0 human checkpoints



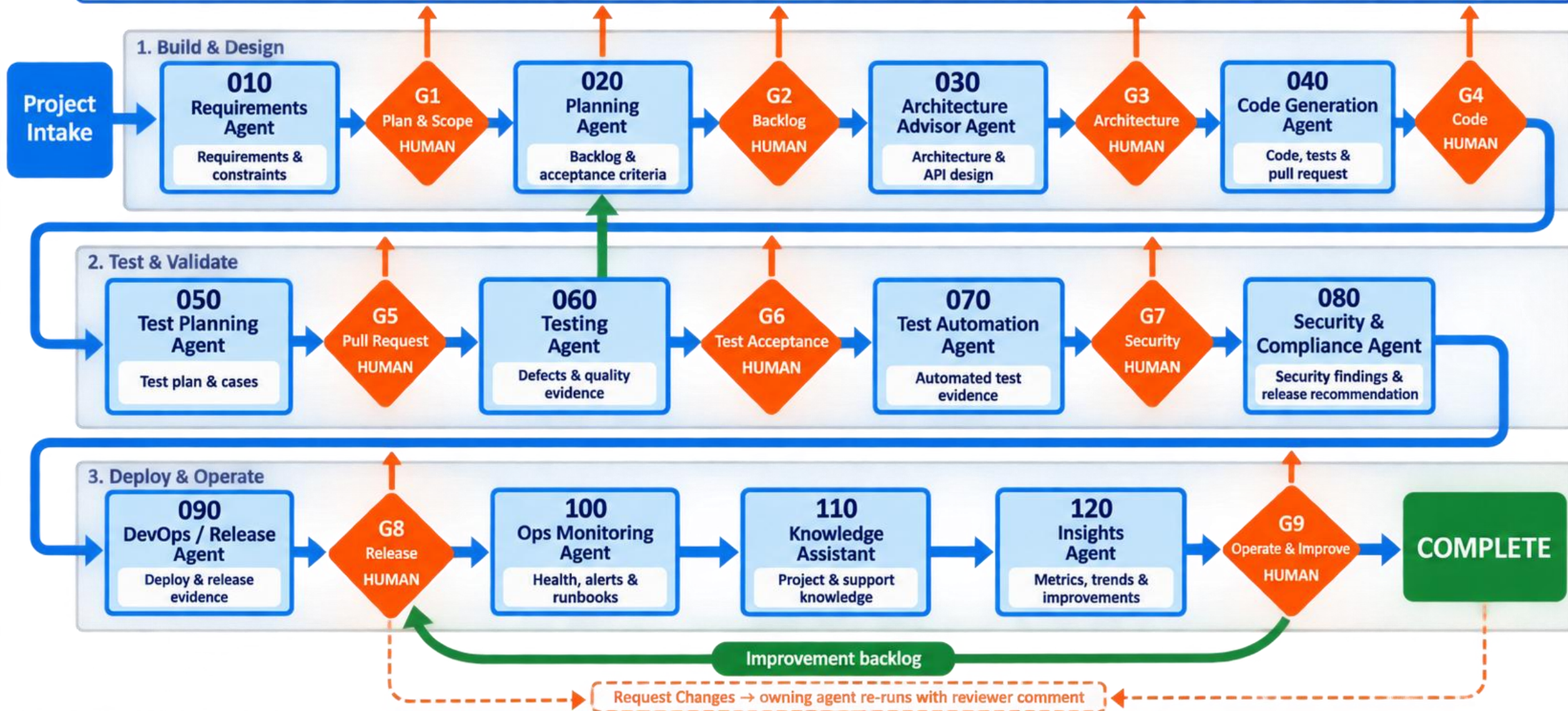
 Evidence validated → gate recorded automatically → next eligible agent starts

Michael Yaacoub | Sr Solution Engineer

Agentic SDLC — Human Review Workflow

HUMAN REVIEW —
9 checkpoints

Reviewer: Review evidence • Edit proposal • Select artifacts • Approve / Publish • Request Changes • Reject • Delegate



Workflow pauses at every gate → authorized reviewer decides → approved publication → next eligible agent starts

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Production controls covered by the POC

The application adds controls around model choice, external actions, recovery, notifications, and lifecycle management.

Model policy

Per-agent model selection, inheritance, project overrides, and Model Router where appropriate

Identity and authorization

Entra ID, guest or email OTP access, configurable roles, server-side permission checks

Tool and content guardrails

APIM boundary, tool allow-lists, token limits, validation, and refusal handling

Resilience

Persisted workflow tasks, bounded retries, persisted circuit breaker, visible deterministic failures

Traceability

Agent Activity and append-only Audit Trail correlate prompts, models, artifacts, tools and decisions

Lifecycle operations

Owner notifications, post-completion revisions, and controlled project cleanup with retained audit projection

What is persisted and where

Azure DevOps and GitHub remain the durable systems of record; orchestration state, telemetry, and runtime assets support execution.

Azure DevOps

Epics through Tasks, sprints, queries, dashboards, delivery plans, Test Plans/Runs, wiki and evidence links

GitHub

Generated application repository, branch, pull request, commits, GitHub Actions and documentation

Cosmos DB

Workflow tasks, checkpoints, agent context, decisions, approvals, operation state and idempotency records

Application telemetry

Agent traces, workflow events, user actions, approvals, audit evidence, metrics and logs

Azure App Service

Published Angular/Ionic UI and FastAPI service, health endpoint and OpenAPI surface

Choose the right execution policy

The topology and controls remain consistent; the policy changes who decides at each gate.

Autonomous

0 human checkpoints

Policy validates evidence and records gate decisions

Best fit

Trusted repeatable patterns, demos, high-volume project sets

Minimal Human Review

4 consequential checkpoints

People review backlog, architecture, pull request, and release

Best fit

Enterprise balance between speed and control

Human Review

All 9 checkpoints

A named reviewer decides every gated transition

Best fit

Regulated, high-risk, unfamiliar, or first-of-kind work

Agentic delivery with control built in

Agents generate. Evidence proves. Policy and people decide.

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Documentation and reference architecture

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